OIL PUMP (Continued)

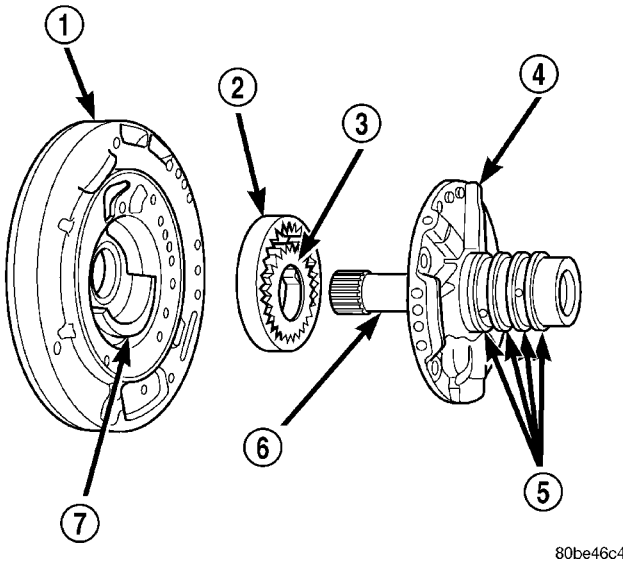


*Fig. 227 Oil Pump Assembly*

1 - PUMP HOUSING
2 - OUTER PUMP GEAR
3 - INNER PUMP GEAR
4 - REACTION SHAFT SUPPORT
5 - SEAL RINGS (4)
6 - REACTION SHAFT
7 - CRESCENT

(3) Remove the pump gears (2, 3) (Fig. 227) and check for wear and damage on pump housing (1) and gears (2, 3).

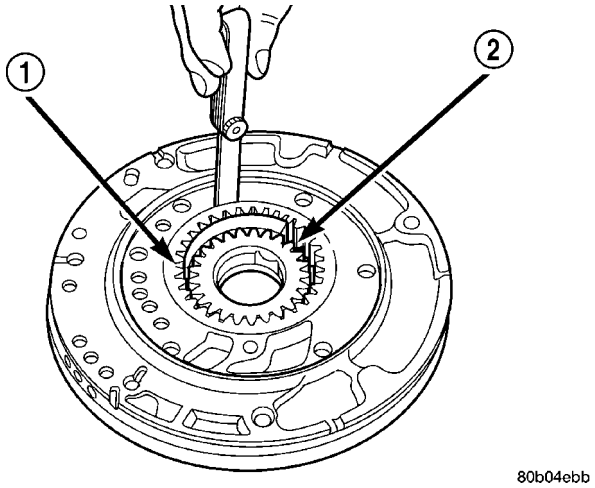


*Fig. 228 Measure Outer Gear to Pocket*

1 - OUTER GEAR
2 - POCKET

(4) Re-install the gears and check clearances.

(5) Measure the clearance between the outer gear (1) and the pump pocket (2) (Fig. 228). Clearance should be 0.089-0.202 mm (0.0035-0.0079 in.).

(6) Measure clearance between outer gear and crescent. Clearance should be 0.060-0.298 mm (0.0023-0.0117 in.).

(7) Measure clearance between inner gear and crescent. Clearance should be 0.093-0.385 mm (0.0036-0.0151 in.).

(8) Position an appropriate piece of Plastigage across both pump gears.

(9) Align the Plastigage to a flat area on the reaction shaft support housing.

(10) Install the reaction shaft to the pump housing. Tighten the bolts to 27 N·m (20 ft. lbs.).

(11) Remove bolts and carefully separate the housings. Measure the Plastigage following the instructions supplied.

(12) Clearance between outer gear side and the reaction shaft support should be 0.020-0.046 mm (0.0008-0.0018 in.). Clearance between inner gear side and the reaction shaft support should be 0.020-0.046 mm (0.0008-0.0018 in.).

## ASSEMBLY

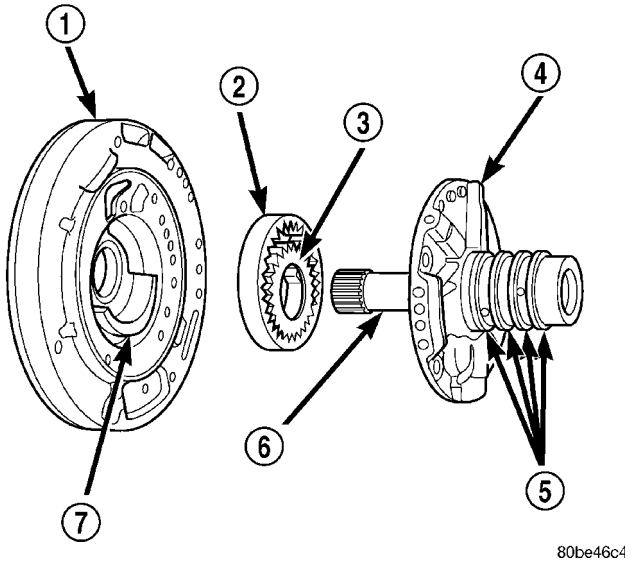


*Fig. 229 Oil Pump Assembly*

1 - PUMP HOUSING
2 - OUTER PUMP GEAR
3 - INNER PUMP GEAR
4 - REACTION SHAFT SUPPORT
5 - SEAL RINGS (4)
6 - REACTION SHAFT
7 - CRESCENT

(1) Assemble oil pump as shown (Fig. 229)

(2) Install and torque reaction shaft support-to-oil pump housing bolts to 28 N·m (20 ft. lbs.) torque.